Choosing variant interpretation tools for clinical applications: context matters

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Abstract

Our inability to solve the variant interpretation problem has become a bottleneck in the biomedical/clinical application of Next-Generation Sequencing.

This situation has favoured the development of several in silico tools. However, choosing the optimal tool for our purposes is difficult because of the high variability of clinical context across and within countries, and over time.

We present a computational procedure, based on the use of cost models, that allows the simultaneous comparison of an arbitrary number of tools across all possible clinical scenarios. We apply our approach to a set of pathogenicity predictors for missense variants, showing how differences in clinical context translate to differences in tool ranking.

Materials and Methods

Variant dataset

We retrieved a set of benign and pathogenic variants from the dbNSFP database, release January 6, 2020. We imposed three filters. First, the variants should not affect a splicing site. Second, the ClinVar review status should be: Practice guideline, Expert Panel or Criteria provided, multiple submitters, no conflicts. Third, we unified the clinical significance classes Benign and Likely_benign as Benign; and Pathogenic and Likely_pathogenic as Pathogenic. The resulting dataset has 11093 variants, 3752 pathogenic and 7341 benign, from 1914 proteins.

Pathogenicity predictors

We have used sixteen pathogenicity predictors: PolyPhen2-HDIV, PolyPhen2-HVAR, SIFT, CADD, MutationTaster2, MutationAssessor, REVEL, LRT, PROVEAN, MetaLR, MetaSVM, VEST, MutPred, PON-P2, PMut, Eve. We compute:

- sensivity (se): True Positive / (True Positive + False Negative)
- specificity (sp): True Negative / (True Negative + False Positive)
- coverage (α): Predicted variants / Total variants

Cost computations

The computations associated with the two cost models used in this work, MISC and MISC+NOA (see below), are coded into separate Python scripts, CSP-noco and CSP-co, respectively. The code is freely available at:
https://github.com/ClinicalTranslationalBioinformatics/clinical_space_partition

Results

Choosing a variant interpretation tools without rejection

This model is appropriate when the consequences of rejection are negligible. For example, when working with proteins for which pathogenicity predictors have very good coverage. the normalized average cost of misclassification errors is:

$$rc = [(1-\rho)(1-sp) - \rho(1-se)]rc1 + \rho\,(1-se)$$

where rc1 is the normalized cost of misclassifying benign variants as pathogenic. The equivalent cost for the misclassification of pathogenic variants as benign is implicit in the formula. ρ is the proportion of pathogenic variants in the genome region sequenced. It is comprised between 0 and 1, and varies with the size and location of the sequenced region. Finally, se and sp are the sensitivity and specificity of the predictor, respectively.

Thus, to compare an arbitrary number of predictors and finding the best of them in terms of cost for a single clinical scenario, the procedure is to compute the rc for each method using the equation and select the method with the lowest rc.

Furthermore, we can extend the previous procedure to all possible clinical contexts varying rc1 between 0 and 1 systematically, and obtaining the cost-optimal predictor at each region. For this purpose, we have developed the algorithm CSP-norej. In Figure 1, we illustrate the application of the model to the comparison of a set of pathogenicity predictors across the whole clinical space:

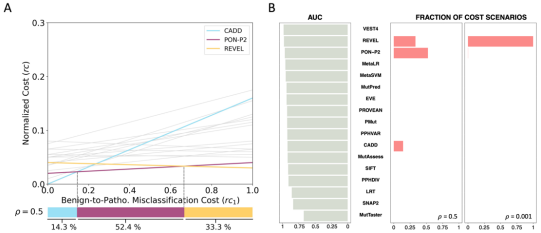

Figure 1 | ***Application of the cost model without rejection to sixteen pathogenicity predictors***
a| Each predictor is represented by a line in which each point corresponds to the normalized cost of the predictor at a given value of the benign-to-pathogenic missclassification error. **b|** Ranking of the sixteen pathogenicity predictors according to AUC. On the right, fraction of cost scenarios for which each predictor is optimal for ρ = 0.5 and 0.001.

Choosing a variant interpretation tools with rejection

We have extended the cost model to include tools with rejection, that is, when predictors do not produce an output. The fraction of cases for which this happens is (1-α), where α is the coverage of the method. In this new scenario, rc is:

$$rc = [\alpha\rho(1-se) + \alpha - 1]\,rc0 + [\alpha(1-\rho)(1-sp) + \alpha - 1]rc1 + 1 - \alpha$$

where rc0, rc1, and ρ are the same as in the previous section. α is the coverage of the method. This can be used to systematically rank predictors in multiple clinical scenarios. For that, we have developed the CSP-rej algorithm that divides the clinical space into regions within which only a single predictor is cost-optimal.

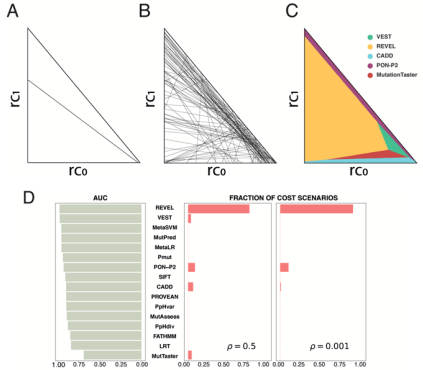

Figure 2 | ***Application of the cost model with rejection to sixteen pathogenicity predictors***
a| Comparison of two predictors. The clinical space is represented by a triangle encompassing all cost scenarios (rc0, rc1). The line divides the triangle into two regions within which a single method is cost-optimal. **b|** As we add more predictors, a pattern of polygons forms; inside each a single predictor is cost-optimal. **c|** Unification of the polygons in regions corresponding to the same predictor. **d|** Ranking of the 16 predictors according to AUC. On the right, fraction of cost scenarios for which each predictor is optimal for ρ = 0.5 and 0.001.

Conclusions

In summary, we have developed a set of algorithms to assess the cost of misclassified errors given a clinical scenario and choose from the diversity of in silico tools, the one more adequate in terms of cost, given their sensitivity, specificity and coverage.

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